PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	The relationship between serum insulin-like growth factor 1 levels and ischemic stroke: a systematic review and meta-analysis
AUTHORS	Li, Yangni; Yang, Weinan; Li, Jinhui; Zhang, Yishu; Zhang, Lincheng; Chen, Simiao; Lan, He; Zhang, Yuyan

VERSION 1 – REVIEW

REVIEWER	Sadeghi, Masoud Kermanshah University of Medical Sciences, Medical Biology Research Center
REVIEW RETURNED	14-Nov-2020

GENERAL COMMENTS	1. There is high heterogeneity in analyses and the authors should use meta-regression. 2. There are grammatical errors. 3. Please add the details about the Stata software used in this meta-analysis. 4. What is the clinical significance of this study? 5. Please arrange studies in figures chronologically (old year to new year from up to down).

REVIEWER	Kanazawa, Masato Niigata University, Neurology
REVIEW RETURNED	16-Nov-2020

GENERAL COMMENTS	Summary: In this systemic review and meta-analysis, authors thoroughly performed the meta-analysis to evaluate the association of insulin-like growth factor 1 (IGF-1) with the risk of ischemic stroke and outcome after ischemic stroke. This meta-analysis showed that higher serum IGF-1 was significantly correlated with less risk of ischemic stroke. However, it could not confirm whether the level of IGF-1 predict outcome after ischemic stroke because of less data. This is a very interesting and important topic. Overall this manuscript is well-written, and following PRISMA checklist, although this reviewer has major concerns. 1. The authors searched China National Knowledge Infrastructure. Publications are not limited to English. Why did the authors search major database and limited to English publications. That made selection bias. Maybe, Begg's test was not shown the publication bias. However, P value was almost 0.1. Funnel plot was slightly deviated. According to this concern, the reviewer was afraid of selection bias. At least, the publications from China National Knowledge Infrastructure were omitted.

	2. In table 3, total number of studies in item, country, sample size, sex, and time was not 17. Were there mixed studies including the list?

REVIEWER	Chinchilli, Vernon M. The Pennsylvania State University, Public Health Sciences
REVIEW RETURNED	06-Dec-2020

GENERAL COMMENTS	There are 17 included studies, yet the authors state that there was an insufficient number of studies to perform Begg's test of publication bias. On page 6 the authors indicate that a study was excluded if it involved more than 1000 cases. It is not clear why the authors would impose such an exclusion criterion. At the bottom of page 6, the authors state that they consider $p < 0.10$ to be statistically significant. The norm is $p < 0.05$. On line 31 of page 7, the authors refer the reader to Table 2 on the quality of the studies, but Table 2 does not appear in the manuscript. Related to that, the authors should state how they assessed the quality of the studies. The authors need to explain much better the outcome variable that is measured in the studies, its units of measurement, and what is a clinically meaningful effect. There are numerous grammatical mistakes, some of which are noted below. The authors should seek editorial assistance from a native English speaker. In lines 55 and 59 on page 4, "evidences" should be "evidence" In the last line on page 4, "uncovered" should be "uncover" In line 7 on page 5, "protective" should be "protective effect" In line 14 on page 5, "correlated to" should be "correlated with" In line 10 on page 8, "paralleled" should be "parallel"

VERSION 1 – AUTHOR RESPONSE

Thanks for your letter and for the editors' comments concerning our manuscript ID bmjopen-2020-045776, entitled "The relationship between serum insulin-like growth factor 1 levels and ischemic stroke: a systematic review and meta-analysis." Those comments are all valuable and very helpful for revising and improving our manuscript, as well as the guiding significance to our researches. We have studied comments carefully and have made correction which we hope to meet with approval. Revised portion are marked in red in the manuscript. The main corrections in the manuscript and the responses to the editors' comments are as flowing:

1. **Comment:** ***Please improve the quality of English throughout.***

Response:

We have read over the full manuscript, sought the help of English professionals and modified it carefully. The revision was marked with red in the manuscript. If there is still a question whether your company provides English editing services, we can pay the relevant fees.

1. **Comment:** ***Please update your search so it is more current (as it was run in Sept 2019).***

Response:

We have updated search to December 29, 2020. “Neuroprotection by Insulin-like Growth Factor-1 in Rats with Ischemic Stroke is Associated with Microglial Changes and a Reduction in Neuroinflammation” was retrieved on PubMed. This article was excluded because the animal experiment did not meet the inclusion criteria; In addition, we conducted a search on Scopus, Embase, China National Knowledge Infrastructure and Web of Science, and found that related articles did not meet the criteria for inclusion.

3. Comment: ***Please include, as a supplementary file, the precise search strategy (or strategies) for one database, including any limits and filters.***

Response:

In the search strategy of the manuscript, we provide a detailed search strategy. In addition, we also provide a supplementary description of the search strategy in additional documents.

4. Comment: ***Please provide side headings in the Discussion to guide the reader including clinical implications of their findings.***

Response:

We added a subtitle to the discussion and annotated it for readers to refer to.

5. Comment: ***Please work to improve the quality of the English throughout your manuscript. We recommend asking a native English speaking colleague to assist you or to enlist the help of a professional copyediting service.***

Response:

We have read over the full manuscript, sought the help of English professionals and modified it carefully. The revision was marked with red in the manuscript. If there is still a question whether your company provides English editing services, we can pay the relevant fees.

6. Comment: ***Please revise the ‘Strengths and limitations’ section of your manuscript (after the abstract). This section should contain five short bullet points, no longer than one sentence each, that relate specifically to the methods. The results of the study should not be summarised here.***

Response:

We have revised the advantages and limitations in the manuscript and marked them.

7. Comment: ***Please ensure that you have fully discussed the methodological limitations of the study in the discussion section of the main text. What is the clinical significance of this study?***

Response:

In the discussion section of the stengths and limitations, we described the shortcomings of the research method in detail. “It provides a basis for in-depth clinical research on the relationship between the level of IGF-1 and neurological score [30] of stroke patients, post-stroke depression [45] dementi [46] and so on. In addition, it is also of great significance to the prevention of stroke (identification of high-risk groups and chemoprevention), early diagnosis, clinical treatment and prognosis.” was written in the manuscript.

8. Comment: ***Please state any competing interests or state ‘None declared’: None declared.***

Response:

On page 10 of the manuscript, we have stated competing interests: None declared.

9. Comment: ***There is high heterogeneity in analyses and the authors should use meta-regression.***

Response:

We have performed a meta-regression analysis and marked on the page 7 of the manuscript.

10. Comment: ***There are grammatical errors.***

Response:

We have read over the full manuscript, sought the help of English professionals and modified it carefully. The revision was marked with red in the manuscript. If there is still a question whether your company provides English editing services, we can pay the relevant fees.

11. Comment: ***Please add the details about the Stata software used in this meta-analysis.***

Response:

We have added the details of the Stata software analysis to the data analysis of the manuscript and marked it.

12. Comment: ***What is the clinical significance of this study?***

Response:

In the discussion section of the stengths and limitations, we described the clinical significance of this study. “In addition, it is also of great significance to the prevention of stroke (identification of high-risk groups and chemoprevention), early diagnosis, clinical treatment and prognosis.” was written in the manuscript.

13. Comment: ***Please arrange studies in figures chronologically (old year to new year from up to down).***

Response:

We have arranged studies in figures chronologically (old year to new year from up to down). It is reflected in table1 and table2 on page 15-19 of the manuscript.

14. Comment: ***Please state any competing interests or state 'None declared': None***

Response:

On page 10 of the manuscript, we have stated competing interests: None declared.

15. Comment: ***The authors searched China National Knowledge Infrastructure. Publications are not limited to English. Why did the authors search major database and limited to English publications. That made selection bias. Maybe, Begg's test was not shown the publication bias. However, P value was almost 0.1. Funnel plot was slightly deviated. According to this concern, the reviewer was afraid of selection bias. At least, the publications from China National Knowledge Infrastructure were omitted.***

Response:

We have considered that if the Chinese data is deleted, the full text data would be too small and the credibility of the conclusion would decrease. Therefore, in order to increase the amount of data, we have added Chinese studies' data.

16. Comment: ***In table 3, total number of studies in item, country, sample size, sex, and time was not 17. Were there mixed studies including the list?***

Response:

In the subgroup analysis, Armbrust et al. (2017) and Aberg et al. (2019) lack data on death and prognosis, so they were not included in the analysis. So the total number of studies in country and sample size is 15. In addition, the total number of sex is 14 because study Peng et al. (2009) has no Male data. As for the subgroup analysis of the time from symptom onset to blood collection, excluding the unknown group Fu et al. (2001), Zhang et al. (2004), Johnsen et al. (2005) and Zhang et al. (2016). There are 5 studies that are ≤24 hours, a total of 4 studies of >48 hours and a total of 7 studies of ≤48 hours which included the study of ≤24 hours.

17. Comment: ***There are 17 included studies, yet the authors state that there was an insufficient number of studies to perform Begg's test of publication bias.***

Response:

The begg test can only be performed on more than 10 studies. The begg test mentioned in the discussion on page 10 is about the relationship between serum IGF-1 and poststroke functional independence. Only 4 studies have relevant data so the begg test cannot be performed. On page 7, we performed the begg test for the total study, including 15 studies which have excluded Armbrust et al. (2017) and Aberg et al. (2019) studies for the lack of prognosis and death-related data.

18. Comment: ***On page 6 the authors indicate that a study was excluded if it involved more than 1000 cases. It is not clear why the authors would impose such an exclusion criterion.***

Response:

We have mentioned it in the discussion: the eligibility criteria that study populations were not less than 25 cases and not more than 1000 cases to prevent large samples from affecting the overall results.

19. Comment: ***At the bottom of page 6, the authors state that they consider p < 0.10 to be statistically significant. The norm is p < 0.05.***

Response:

We have revised to p < 0.05.

20. Comment: ***On line 31 of page 7, the authors refer the reader to Table 2 on the quality of the studies, but Table 2 does not appear in the manuscript. Related to that, the authors should state how they assessed the quality of the studies.***

Response:

Table 2 is at the bottom 19 pages of the manuscript.

Table 2 Newcastle-Ottawa Scale for quality assessment

Study	Exposed cohort	Non-exposed cohort	Ascertainment of exposure	Outcome of interest	Control for factor	Assessment of outcome	Follow-up long enough	Adequacy of follow-up	Total score
Fu et al. (2001) [22]	*	*	*		**	*			6
Sun et al. (2001) [27]	*	*	*		**	*			6
Liu et al. (2002) [25]	*	*	*		**	*			6
Denti et al. (2004) [10]	*	*	*		**	*	*	*	8
Zhang et al. (2004) [29]	*	*	*		**	*			6
Johnsen et al. (2005)	*	*	*		**	*			6

[12]									
Geng et al. (2006) [23]	*	*	*		**	*	*		7
Peng et al. (2009) [26]	*	*	*		**	*			6
Chen et al. (2010) [21]	*	*	*		**	*	*		7
Aberg et al. (2011) [14]	*	*	*		**	*	*	*	8
Dong et al. (2014) [11]	*	*	*		**	*			6
Tang et al. (2014) [13]	*	*	*		**	*	*	*	8
Zhang et al. (2016) [30]	*	*	*		**	*	*	*	8
Li et al. (2016) [24]	*	*	*		**	*			6
Armbrust et al. (2017) [19]	*	*	*		**	*	*	*	8

Tao et al. (2018) [28]	*	*	*		**	*	*	*	8
Aberg et al. (2019) [20]	*	*	*	*	**	*	*	*	9

*The study scored one point in the project; **The study scored two points in the project.

21. Comment: ***The authors need to explain much better the outcome variable that is measured in the studies, its units of measurement, and what is a clinically meaningful effect.***

Response:

We added this piece of content and marked it on the page 5 of the manuscript.

22. Comment: ***There are numerous grammatical mistakes, some of which are noted below. The authors should seek editorial assistance from a native English speaker.***

Response:

We have corrected the grammatical error you raised. We have read over the full manuscript, sought the help of English professionals and modified it carefully. The revision was marked with red in the manuscript. If there is still a question whether your company provides English editing services, we can pay the relevant fees.

VERSION 2 – REVIEW

REVIEWER	Chinchilli, Vernon M. The Pennsylvania State University, Public Health Sciences
REVIEW RETURNED	05-Mar-2021

GENERAL COMMENTS	MAJOR REMARKS The authors need to include a forest plot of the primary meta-analytic result. MINOR REMARKS The last sentence in the Statistical Analyses section should be removed because it does not make sense (And then select the meta-analysis function to analyze separately.) In the first sentence under Methods, "was carroed out" should be "was carried out" There are other grammatical and typographical errors that the authors should fix.

VERSION 2 – AUTHOR RESPONSE

1.Comment：Please include, as a supplementary file, the precise search strategy (or strategies) for one database, including any limits and filters. This should also be cited in the main text.
Response：
We have made corresponding modifications according to your requirements, added additional files, and screened strictly according to the PICO principle.
2.Comment：The authors need to include a forest plot of the primary meta-analytic result.
Response：
We uploaded the forest map additionally.
3.Comment：The last sentence in the Statistical Analyses section should be removed because it does not make sense (And then select the meta-analysis function to analyze separately.)
Response:
We made the corresponding modification based on your comment.
4.Comment:In the first sentence under Methods, "was carroed out" should be "was carried out"
Response:
It may be due to a grammatical error caused by the file format. After downloading, it is found that the text is " carried out".
5. Comment: There are other grammatical and typographical errors that the authors should fix.
Response:
We checked the full text again and marked the revised law.
6. Commnet: Competing interests of Reviewer: None declared.
Response:
We made the corresponding modification based on your comment.

VERSION 3 – REVIEW

REVIEWER	Chinchilli, Vernon M. The Pennsylvania State University, Public Health Sciences
REVIEW RETURNED	18-Jan-2022

GENERAL COMMENTS	In the Methods section of the ABSTRACT, "was carried out eligible studies" should be "was carried out in eligible studies" In the second bullet point under Strengths and Limitations section, "include studies" should be "included studies" In the middle of page 4, "dissminated" should be "disseminated" In the middle of page 4, "artical" should be "article" At the bottom of page 4, "fifield" should be "field" At the top of page 5, the authors state that the "Study populations were not less than 25 cases and no more than 1000 cases." This is not an appropriate criterion for study exclusion – the authors should mention how many studies, if any, were excluded due to this criterion. If there are any such studies, then the authors should perform a sensitivity analysis that includes these studies. Under Statistical Analysis on page 6, "which manifested a publication bias" should be "which manifests a publication bias"

	Table 1 should include the study design, i.e., case-control or cohort. If the majority are cohort studies, then it would be more appropriate to estimate the impact of risk ratios and the impact of IGF-1 on them. Table 4 investigates heterogeneity within subgroups and is not that interesting. A more interesting table would consist of subgroup analyses that compares IGF-1 levels between stroke and non-stroke groups. On page 10, “consisitent” should be “consistent” On page 11, “dementi” should be “dementia” The authors should seek professional help with their grammar.

VERSION 3 – AUTHOR RESPONSE

1.Comment：At the top of page 5, the authors state that the “Study populations were not less than 25 cases and no more than 1000 cases.” This is not an appropriate criterion for study exclusion – the authors should mention how many studies, if any, were excluded due to this criterion. If there are any such studies, then the authors should perform a sensitivity analysis that includes these studies.

Response：

Thank you very much for your valuable comments. First, we set eligibility criteria that study populations were not less than 25 cases and not more than 1000 cases to prevent large samples from affecting the overall results. In addition, we have explained the reasons for the exclusion of documents from each database search in Figure 1. We found that there are no studies where the sample size is not matched but all other criteria are met.

2.Comment：Table 4 investigates heterogeneity within subgroups and is not that interesting. A more interesting table would consist of subgroup analyses that compares IGF-1 levels between stroke and non-stroke groups.

Response：

Thank you very much for your valuable comments. First, this meta-analysis aims to assess the association of insulin-like growth factor 1 (IGF-1) with the risk of incident ischemic stroke and the outcome after ischemic stroke. The Mean, SD, Total of serum IGF-1 (ng/mL) in case(stroke) and control(non-stroke) groups is a main indicator of stroke risk assessment. When we did subgroup analysis, we selected country, sample size, etc. as variables, while IGF-1 levels are only a main indicator of stroke risk assessment.

3.Comment：Table 1 should include the study design, i.e., case-control or cohort. If the majority are cohort studies, then it would be more appropriate to estimate the impact of risk ratios and the impact of IGF-1 on them.

Response：

Thank you very much for your valuable comments. In manuscript eligibility criteria (Page 5), we mention that study used a prospective or retrospective cohort design, or nested case-control design.

3.Comment：Please work to improve the quality of the English throughout your manuscript.

Response：

Thank you very much for your valuable comments. We have corrected the specific error you raised. We also checked the full text and fixed some grammatical errors. If there are still grammar problems, whether your company can provide English editing services, we can pay the relevant fees.

VERSION 4 – REVIEW

REVIEWER	Chinchilli, Vernon M. The Pennsylvania State University, Public Health Sciences
REVIEW RETURNED	19-Feb-2022

GENERAL COMMENTS	In line 43 on page 4, “the the” should be “the” The authors should mention if any studies were excluded due to having fewer than 25 cases or more than 1000 cases. If there are such studies, then the authors should include them in a sensitivity analysis. In the Statistical Analyses section, the authors should mention that they applied a meta-regression. In line 43 on page 6, “tool..” should be “tool.” In line 45 on page 6 “country,),” should be “country),” In line 54 on page 7, “correlation” should be “difference” In the last line on page 7, “no significant difference” should be “a significant difference”

VERSION 4 – AUTHOR RESPONSE

1.Comment：In line 43 on page 4, “the the” should be “the”
Response：
Thank you very much for your valuable comments. We have revised and annotated the manuscript.
2.Comment：The authors should mention if any studies were excluded due to having fewer than 25 cases or more than 1000 cases. If there are such studies, then the authors should include them in a sensitivity analysis.
Response：
Thank you very much for your valuable comments. You also mentioned this in your previous rework comment. We may not have explained it clearly before, so now we will give a more precise explanation. First, we have established Eligibility criteria, and only studies that fulfilled the criteria can be included. Therefore, we found 88 full-text articles assessed for eligibiliity. According to the evaluation mechanism, a total of 60 full-text articles were excluded for the following reasongs, 14 were reviews, 19 had no human experiments, 17 had no relevant results, 3 were protocol papers and 7 lacked useable data. So none studies were excluded due to having fewer than 25 cases or more than 1000 cases. Finally, we have explained the reasons for the exclusion of documents from each database search in Figure 1.
3.Comment：In the Statistical Analyses section, the authors should mention that they applied a meta-regression.
Response：
Thank you very much for your valuable comments. We have added meta-regression analysis in the Statistical Analyses section.
4.Comment：In line 43 on page 6, “tool..” should be “tool.”
Response：

Thank you very much for your valuable comments. We have revised and annotated the manuscript.

5.Comment：In line 45 on page 6 “country,),” should be “country),”

Response：

Thank you very much for your valuable comments. We have revised and annotated the manuscript.

6.Comment：In line 54 on page 7, “correlation” should be “difference”

Response：

Thank you very much for your valuable comments. We have revised and annotated the manuscript.

7.Comment：In the last line on page 7, “no significant difference” should be “a significant difference”

Response：

Thank you very much for your valuable comments. We have revised and annotated the manuscript.